

AI-Powered PCAP Analysis & Threat Detection

An AI-assisted cybersecurity investigation approach that transforms PCAP network traffic into structured data, identifies investigation-relevant patterns, validates findings statistically, develops detection rules, and supports evidence-based incident response.

TOOLS & TECHNOLOGIES

Wireshark • TShark • Python • Excel • Generative AI • Statistical Analysis • Data Visualization

ORIGINAL PCAP

2021-08-16-formbook.pcap

3,588 packets / records



COMBINED PCAP-DERIVED TRAFFIC CSV

Combined_Traffic_Data.csv

3,588 traffic records

PROJECT FOCUS

*Evidence-based investigation: AI-assisted analysis,
independently verified before any security conclusion.*

Investigation Overview

A SOC scenario involved captured network traffic requiring rapid security analysis. Traditional PCAP investigation using tools such as Wireshark can require significant manual effort to examine thousands of packets. This project demonstrates how Generative AI can accelerate traffic analysis by identifying patterns and investigation priorities, while statistical and analyst verification are used to validate the results.



3,588

Total Records



3,549

IP-Bearing Records



10.10.0.33

Primary Investigation Host

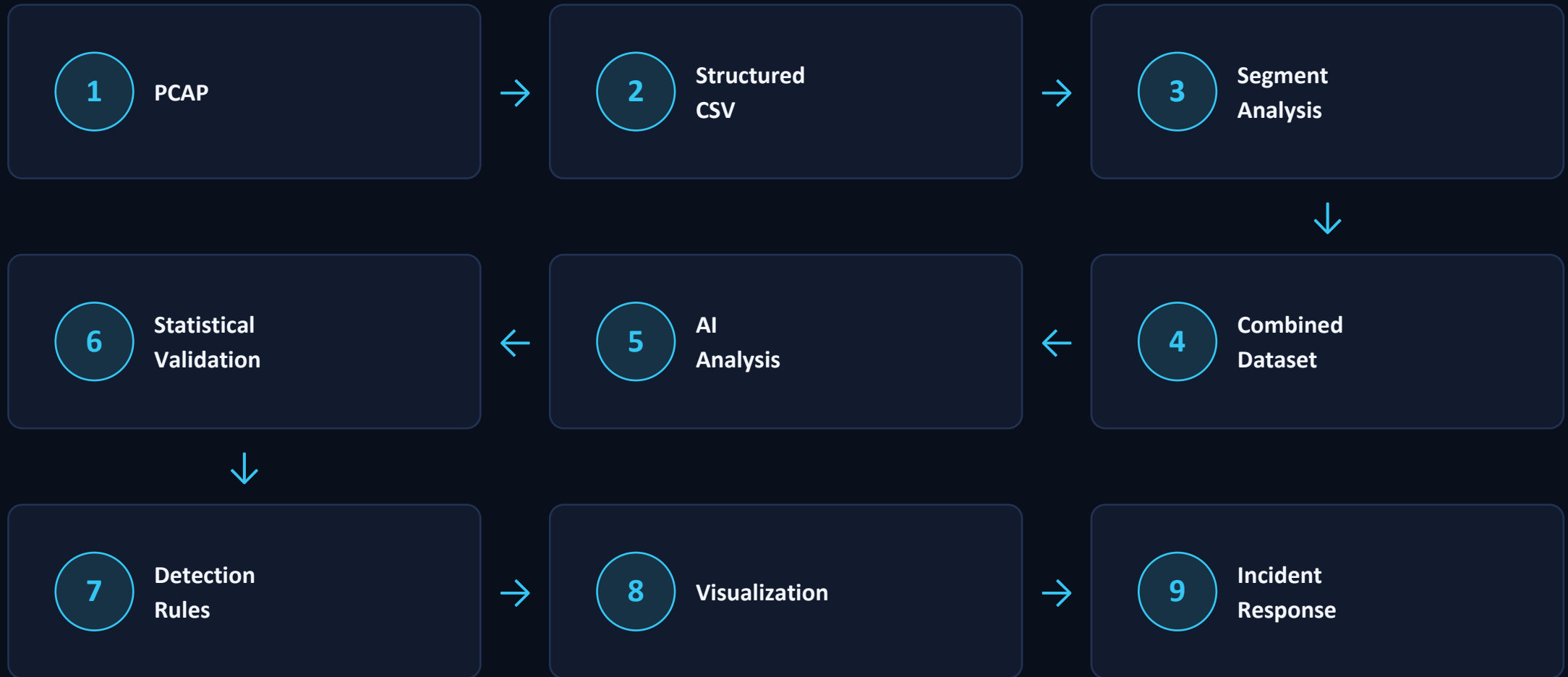


10 min 10.40 sec

Capture Duration

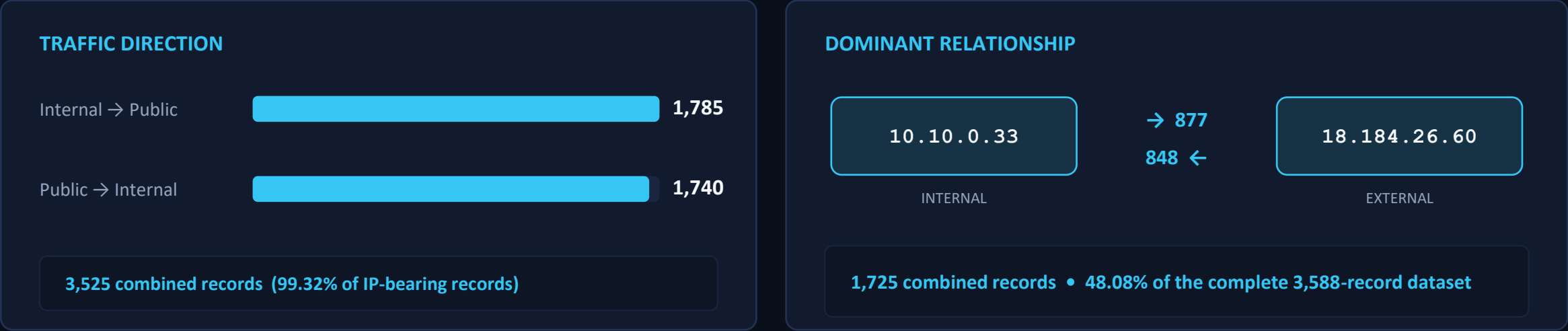
“AI accelerates investigation; analyst verification establishes confidence in the findings.”

Investigation Workflow



PCAP-derived traffic was transformed into structured data, analysed using AI-assisted methods, independently validated, converted into detection logic and visual evidence, and finally used to support an evidence-based incident response strategy.

Key Network Traffic Findings



The remaining 24 IP-bearing records: 16 local broadcast records (10.10.0.255) and 8 multicast records (239.255.255.250). Excluded from the Internal ↔ Public calculation as special local/network-discovery traffic, not ordinary public-unicast communication.

Communication concentration is investigation-relevant, but volume alone does not establish malicious activity.

Investigation-Relevant Indicators

TCP/53480

804

records

ALL 804: 18.184.26.60:80 → 10.10.0.33:53480

Protocol composition: 803 TCP-only • 1 with http:media decoding

Port number alone is not proof of malicious activity.

DNS

112

records

58: 10.10.0.33 → 8.8.8.8

54: 8.8.8.8 → 10.10.0.33

Requires contextual validation

FRAME SIZE

4

statistical outliers

52,160 • 53,529 • 54,032 • 58,337 bytes

IQR upper threshold: 3,695 bytes

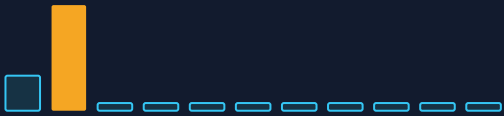
Frame size alone does not establish exfiltration, tunnelling, or compromise.

TRAFFIC BURST

2,097

records • peak 01:59–02:00

Capture duration: ~10 min 10.40 sec (10.1733 min)



A volume spike alone does not establish malicious behaviour.

Repeated and bursty communication was observed. No stable periodic interval was established.

BEACONING: NOT CONFIRMED

Large DNS-associated frames do not independently prove DNS tunnelling or exfiltration.

Detection Rules & AI-Assisted Analysis

FIVE DETECTION RULES

1

Possible Automated External Communication Detection

2

Unencrypted HTTP Communication Monitoring

3

Unusual Data Transfer Volume Detection

4

DNS Traffic Anomaly Detection

5

New External Communication Monitoring

AI CONTRIBUTION

- ▶ Identifying communication patterns
- ▶ Classifying internal / external relationships
- ▶ Examining protocol and port distributions
- ▶ Identifying investigation priorities
- ▶ Interpreting visualizations
- ▶ Developing detection-rule logic
- ▶ Supporting incident-response reasoning

HUMAN / ANALYST VERIFICATION

Human verification validated record counts, communication relationships, IP classification, protocol classification, port observations, frame-size statistics, timeline results, traffic direction, and final security interpretation.

“AI was used as an analytical assistant — not as final evidence.”

Incident Response Strategy



“Containment, eradication and recovery should be proportional to validated evidence and should not assume compromise without confirmation.”

Final Security Assessment & Recommendations

CURRENT STATUS

INVESTIGATION REQUIRED

COMPROMISE STATUS

NO CONFIRMED COMPROMISE

OBSERVED

- 3,525 Internal ↔ Public records
- 1,725 dominant relationship (10.10.0.33 ↔ 18.184.26.60)
- 804 TCP/53480 records
- 112 DNS records
- 4 frame-size outliers
- 2,097-record peak (01:59–02:00)

NOT ESTABLISHED

From available packet metadata:

- ✗ Malware
- ✗ C2 communication
- ✗ Data exfiltration
- ✗ DNS tunnelling
- ✗ Attacker identity
- ✗ Malicious intent
- ✗ Beaconing (not confirmed)

RECOMMENDED ACTIONS

- 1 Prioritize 10.10.0.33
- 2 Validate 18.184.26.60 communication
- 3 Review endpoint processes/connections
- 4 Investigate DNS queries and domains
- 5 Correlate security/system logs
- 6 Use approved threat intelligence
- 7 Compare against known-good baseline
- 8 Continue heightened monitoring

Investigation-relevant activity was identified, but confirmed compromise was not established from available packet metadata alone.

INVESTIGATION
REQUIRED

Lessons Learned & Conclusion



LESSON 1

Data Validation

Reliable analysis depends on correctly structured, validated traffic data.



LESSON 2

AI Accelerates Analysis

AI helped identify patterns, relationships, and investigation priorities efficiently.



LESSON 3

Verify AI Results

AI observations must be checked against the dataset and statistical analysis.



LESSON 4

Unusual ≠ Malicious

Unusual behaviour must be validated before being classified as an attack.



LESSON 5

Visualization Helps

Visuals make complex relationships, anomalies, and timelines easier to understand.

“AI can accelerate cybersecurity investigation, but analyst verification and additional evidence are required before confirming compromise.”

FINAL TAKEAWAY: INVESTIGATION REQUIRED